

M. J. Biopharm Private Limited
Plot No. 18, International Biotech Park, Hinjawadi, Phase II, Pune 411057.

**Format for Attachment**

Title: Preliminary Investigation for Deviation occurred during calibration of TOC Analyzer
(Instrument ID: I/QC/018)

Attachment No: VIII

Ref. document No: DEV/QC/25/010

Preliminary Investigation:

Monthly scheduled calibration of TOC Analyzer (Instrument ID: I/QC/018) was due on 21/03/2026, accordingly calibration of the TOC Analyzer, planned on 21/03/2026 and work allotted to the analyst (Emp. ID: 615).

As per SOP 'Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L CPH with LabSolutions software' (SOP/DP/QC/045-R02), 2- point calibration & System suitability test needs to be performed during monthly calibration.

Therefore, as per plan, prior to initiation of calibration, analyst (Emp. ID: 615) has assigned A.R. No. NR/25/0438 for the calibration activity in accordance with SOP No. SOP/DP/QC/008 and verified all the pre requisites before initiation of 2-Point calibration and System suitability test as per SOP, 'Title: Operation, Calibration and Maintenance of Shimadzu TOC Analyzer Model No.: TOC-L CPH With LabSolutions Software, SOP No.: SOP/DP/QC/045.

As per SOP, analyst injected the blank water to check the TOC, and obtained result (79.78 ppb) was found well within the defined acceptance criteria. The SOP specifies that the TOC of blank water should be Not More Than 100 ppb. Refer attachment I.

Since, TOC of blank water, found within the limit, therefore analyst performed the 2-point calibration to generate the calibration curve and details of the generated curve as below; Refer Attachment II

Name of Curve: CAL_21032026.2026_03_21_11_54_09.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	5.432	0.1581	2.91 %
Two Point Calibration (500 ppb)	42.58	0.4564	1.07 %

Further calculated the TOC of blank water which is injected to establish the calibration curve and observed that obtained TOC of blank water is 73 ppb, which was within the acceptance criteria (NMT 100 ppb).

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**Format for Attachment**

Since, obtained results for two-point calibration found well within the limit as per SOP/DP/QC/045, hence analyst proceeded for 'System Suitability Test' (SST), however after completion of the SST, it was observed that obtained SST results i.e. 115.1 % does not meet the acceptance criteria (Limit: 85 % to 115 %). Refer attachment -III).

As per SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L_{CPH} with LabSolutions Software, Section No.: 7.4.3.26, The results of system suitability test must not be less than 85 % & must not more than 115 %, therefore, initiated the deviation for further investigation.

Based on the occurred discrepancy following immediate action has been taken,

- 1) The ongoing calibration activity was immediately stopped after observing that the SST result (% Recovery of Benzoquinone – 115.1%) was outside the acceptance criteria (85.0%–115.0%).
- 2) All glassware which are used for blank and preparation of standard solutions were rechecked and quarantined the analysis tray.
- 3) The "Out of Calibration" Label was affixed on instrument as per SOP/DP/QA/020.
- 4) Verified the 2-point calibration result and noted that result is well within acceptance criteria.
- 5) As the System Suitability Test (SST) results did not meet the acceptance criteria, therefore as an immediate action, it was informed to perform the alternative test i.e. 'Oxidizable Substance' as per purified water standard test procedure i.e. DP/STP/UT/0003 for the scheduled water sample analysis.
- 6) The observed Non-Conformance immediately informed to Team lead and Head QC.

Based on the reported nonconformance, verified the analyst workbench for standard preparation and noted that analyst prepared the sucrose stock standard solution (50000 ppb) in 100 ml volumetric flask and 1,4-Benzoquinone stock standard solution (50000 ppb) in 100 ml volumetric flask. Analyst used amber colored volumetric flask during preparation of 1,4-Benzoquinone standard.

The prepared standard solutions for 2-point calibration were verified and noted that Sucrose stock standard solution (50,000 ppb) was prepared in a 100 mL volumetric flask correctly as per the procedure defined in SOP/DP/QC/045.

The calibration data sheet (Format No. SOP/DP/QC/045/F01 and SOP/DP/QC/045/F02) was verified for the recorded preparation details of Sucrose and 1,4-Benzoquinone standard solutions and noted that, analyst prepared the solutions correctly. (Refer attachment -IV & V), the details are as below;

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**Format for Attachment****Preparation of Sucrose Standard Solution:**

Weighed 11.883 mg of sucrose & transferred in 100 ml volumetric flask and diluted to 100 ml with purified water. (Refer Attachment IV & VI)

Further, taken 1 ml of above stock standard (50000 ppb) by pipetting with Glass pipette in 100 ml volumetric flask and make up the final volume to 100 ml with purified water. (Refer Attachment IV)

Verified the details of sucrose standard which was used for the preparation of 500 ppb solution and noted that analyst used the correct and valid standard, the details are as below;

Name of Standard: Sucrose

Lot No: 4178807, (In-house No.: PS/009/001)

Validity: 17/02/2027

Preparation of 1, 4-Benzoquinone Standard Solution:

Weighed and dissolved 7.499 mg of 1, 4-Benzoquinone USPRS in 100 ml of purified water. (Refer Attachment V & VI)

Further, diluted 1 ml of 1, 4-Benzoquinone stock standard solution (50000 ppb) to 100 ml with purified water to get 500 ppb 1, 4-Benzoquinone.

Verified the details of 1,4 –Benzoquinone standard which is used for the preparation of 500 ppb solution and noted that analyst had used the correct and valid standard, the details are as below; (Refer Attachment V)

Name of Standard: 1,4 –Benzoquinone USPRS

Lot No: R15690 (In house Vial No.: RS/002/23-002)

Validity: Current lot

After preparation of standards analyst performed the 2-point calibration by preparing the calibration curve 'CAL_21032026' by referring the SOP/DP/QC/045 and generated curve 'CAL_21032026.2026_03_21_11_54_09.cald'. Obtained results of the 2-point calibration found well within the acceptance criteria, the details are as follow, (Refer Attachment - II)

After generation of the calibration curve, performed the SST by selecting the latest calibration curve 'CAL_21032026.2026_03_21_11_54_09.cald', however after completion of the SST, it was observed that obtained SST results i.e. 115.1 % does not meet the acceptance criteria (Limit: 85 % to 115 %). Refer attachment -III).

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Format for Attachment		

SST Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
System Suitability Test'	48.19	1.227	2.55 %
Recovery of 1, 4-Benzoquinone (Limit: Between 85-115 %)	115.1 %		

Verified the vial of 1,4 –Benzoquinone USPRS, Lot No: R15690 (In house Vial No.: RS/002/23-002) to check any abnormality with the remaining quantity of standard and noted that, on document remaining quantity was 41.988 mg, however physically the standard quantity almost consumed as it was lost during the handling and remaining quantity not sufficient for further activity.

Since, assignable cause could not be identified during the preliminary investigation, therefore conducted the analyst interview to find out the root cause and as an outcome of interview, it was suspected that the non-conformance was occurred might be due to dilution error which may have occurred inadvertently while preparing the standard solution, leading to a slightly higher recovery of the 1, 4-Benzoquinone. Refer attachment VII (Analyst interview).

Inference:

Based on the outcome of preliminary investigation and analyst interview, assignable cause could not be identified, however suspected cause for the occurred non-conformance could be dilution error which may have occurred inadvertently while preparing the standard solution, leading to a slightly higher recovery of the 1, 4-Benzoquinone, therefore to find out the root cause additional investigation followed by hypothesis testing needs to be performed.

Recommended action I:

Based on the suspected cause, higher side results might be occurred due to improper dilution of the 1,4 – Benzoquinone standard. Hence, to rule out the dilution error it is recommended to perform the re-dilution of 1,4-Benzoquinone stock standard solution (50000 ppb) as below and inject the solution as per below sequence.

Take 1 ml of 1,4-Benzoquinone stock standard solution (50000 ppb) and dilute to 100 ml with purified water.
(1,4 -Benzoquinone Working Standard (500 ppb);

Sr. No.	Name of Sample	No. of Sample to be injected
01	Same Vial	01
02	Refilled Vial	01
03	Re-Diluted Vial	01

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Prepared By (QC) (Sign & Date)	Reviewed By (QC) (Sign & Date)	Reviewed By (QA) (Sign & Date)
 21/03/26	 21/03/26	 21/03/26

21-03-2026 16:10:23(+05:30) : Confirm[Akash Ghadage]Other CommentData submitted for review

21-03-2026 17:56:35(+05:30) : Review[Nilesh Dushing]Other CommentData Reviewed. (Hypothesis testing performed against Deviation No.:DEV/QC/25

Date of creation 2026/03/21 15:27:42
 Sample Run TOC_I-QC-018_2026 - 5-566-3 - CAL_21032026_01.tlxl
 User Akash Ghadage
 Object ID 0L-H54436004105-13525E124CD1-0000
 System TOC-L + ASI-L
 Origin TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald

Sample

Sample Name: 1-4 Benzoquinone (Same Vial)
 Sample ID: Std SST
 Object ID 0L-H54436004105-13525E124CD1-0000
 Origin: TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald
 Status Completed
 Chk. Result 117.0% : USP/EP system suitability test: Fail
 Comment

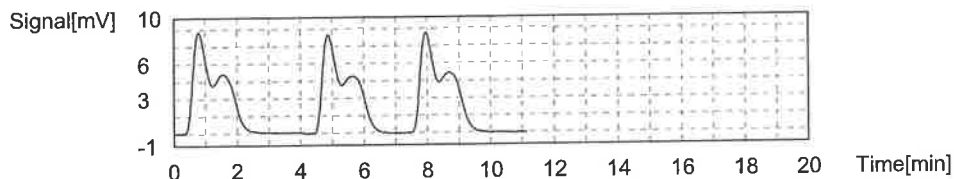
Type	Anal.	Manual Dilution	Result
Unknown	NPOC	1.000	NPOC:658.3ppb

1. Det

Anal.: NPOC

No.	Area	Conc.	Inj. Vol	Aut. Dil	Rem.	Ex.	Cal. Curve	Date / Time
1	49.38	664.6ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:19:52
2	48.17	648.4ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:23:20
3	49.18	661.9ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:27:41

Acid Add. 0.000%
 Spurge Gas Flow 80mL/min
 Sp. Time 60.00sec
 Mean Area 48.91
 Mean Conc. 658.3ppb
 SD Area 0.6486
 CV Area 1.33%
 SD Conc 8.730
 CV Conc 1.33%
 Vial 3
 No Washes 2



History

21-03-2026 16:10:22(+05:30) : Confirm[Akash Ghadage]Other CommentData submitted for review

21-03-2026 17:56:35(+05:30) : Review[Nilesh Dushing]Other CommentData Reviewed. (Hypothesis testing performed against Deviation No.:DEV/QC/25

Date of creation 2026/03/21 15:45:33
Sample Run TOC_I-QC-018_2026 - 5-566-4 - CAL_21032026_01.txl
User Akash Ghadage
Object ID 0L-H54436004105-13525E124CD6-0000
System TOC-L + ASI-L
Origin TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald

Sample

Sample Name: 1-4 Benzoquinone (Refilled Vial)
Sample ID: Std SST
Object ID 0L-H54436004105-13525E124CD6-0000
Origin: TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald
Status Completed
Chk. Result 104.6% : USP/EP system suitability test: Pass
Comment

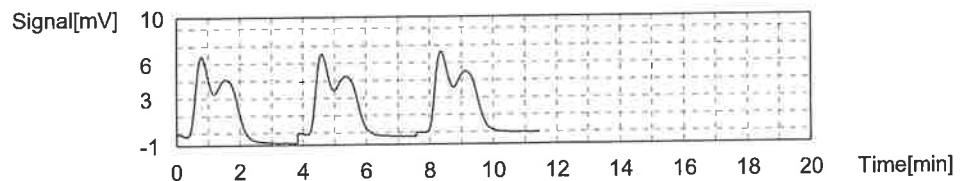
Type	Anal.	Manual Dilution	Result
Unknown	NPOC	1.000	NPOC:595.9ppb

1. Det

Anal.: NPOC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Cal. Curve	Date / Time
1	43.79	589.4ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:37:05
2	43.68	587.9ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:41:16
3	45.35	610.4ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:45:32

Acid Add. 0.000%
Spurge Gas Flow 80mL/min
Sp. Time 60.00sec
Mean Area 44.27
Mean Conc. 595.9ppb
SD Area 0.9340
CV Area 2.11%
SD Conc 12.57
CV Conc 2.11%
Vial 4
No Washes 2

**History**

21-03-2026 16:10:22(+05:30) : Confirm[Akash Ghadage]Other CommentData submitted for review

21-03-2026 17:56:35(+05:30) : Review[Nilesh Dushing]Other CommentData Reviewed. (Hypothesis testing performed against Deviation No.:DEV/QC/25

Date of creation 2026/03/21 16:03:46
Sample Run TOC_I-QC-018_2026 - 5-566-5 - CAL_21032026_01.tlx
User Akash Ghadage
Object ID 0L-H54436004105-13525E124CD6-0001
System TOC-L + ASI-L
Origin TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald

Sample

Sample Name: 1-4 Benzoquinone (Re-Diluted Vial)
Sample ID: Std SST
Object ID 0L-H54436004105-13525E124CD6-0001
Origin: TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald
Status Completed
Chk. Result 76.26% : USP/EP system suitability test: Fail
Comment

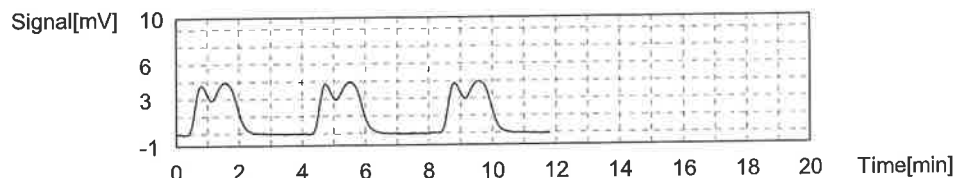
Type	Anal.	Manual Dilution	Result
Unknown	NPOC	1.000	NPOC.454.4ppb

1. Det

Anal.: NPOC

No.	Area	Conc.	Inj. Vol	Aut. Dil	Rem.	Ex.	Cal. Curve	Date / Time
1	33.43	450.0ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:55:04
2	34.08	458.7ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 15:59:33
3	33.77	454.5ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 16:03:45

Acid Add. 0.000%
Spurge Gas Flow 80mL/min
Sp. Time 60.00sec
Mean Area 33.76
Mean Conc. 454.4ppb
SD Area 0.3251
CV Area 0.96%
SD Conc 4.376
CV Conc 0.96%
Vial 5
No Washes 2



History